

P&ID Redline Punch List & BOM Review

Hayfield Ridge Amine Facility — MPLX (Kinder Morgan Treating LP) · Lea County, New Mexico

Rev 1 issued: 2026-08-12 — corrected against the current authoritative redline set (“mplx_redline.pdf”, 40 sheets)

Rev 0: 2026-08-09 (430 items from the earlier field markup set)

Method: every Rev 0 item re-verified mark-by-mark on the current set; duplicates merged; new marks added

1 Executive Summary

40
P&IDS RE-REVIEWED

137
OPEN PUNCH ITEMS

15
CRITICAL ITEMS

292
CLOSED (NOT ON CURRENT SET)

3
DUPLICATES MERGED

2
NEW MARKS ADDED

38
BOM LINE ITEMS

10
ENGINEERING QUESTIONS

This Rev 1 corrects the 2026-08-09 review against the current authoritative redline set supplied 2026-08-12. Every one of the 430 Rev 0 punch items was re-verified individually at high magnification on the new set: **137 remain open** (marks still present), **292 are closed** because their marks no longer appear on the current set, **3 were duplicates** — the same physical work marked on two adjacent sheets — and are merged into single items, and **2 new red-drawn marks** found only on the current set are added (02003-08 , 02004-13). Item keys are unchanged from Rev 0, so any sign-offs already recorded remain valid; closed items are listed in the Disposition Register (Section 7), never silently deleted.

Read before closing anything: the removed marks were deleted from the markup — the base drawings did not change revision, and this review cannot confirm the underlying work is complete. Safety-significant deletions (a backwards check valve note, the 1440-psig relief-routing question, and every “missing PSV” highlight) are flagged in Section 7 for field confirmation at PSSR walkdown.

What remains open is dominated by hand-drawn hardware additions the markup author kept: small-bore vents/drains/bleeds with plugs (~30 valves ½"-2"), the JT-skid regulator set and LCV station valves, the blowcase gauge-drain hardware, KO-drum inlet/drain valves and 14"-line high-point vents, the hot-oil blanket-gas system (PCV-2001 + ¾" run) and burner vent XV assemblies, the FE/FIT installations on both VCU assist-gas lines and the FIT P-3 meter, the slop-tank drain re-pipe (two new 2" heat-traced lines, 6"x4" PVRV transitions, four 1" drains), the air-system relief redesign (PSV removals + vents + gauge relocations — pending Q16), and drawing/ISO corrections.

2 Sheet Index — open items per sheet

01001 Slug Catcher (X-11001 / X-12001) · G — 13	02015C Expansion Tank TK-45001A (Hot Oil · E — 6
01003 Gas Scrubber (V-21001 / V-22001) · E — 7	02015D Combustion Air Blower / Burner BRN · E — 2
02001 Inlet Coalescing Filter S-1100 (40 · F — 5	02016 Catch Tank TK-2001 / TK-2002 (400 · D — 2
02002 Amine Contactor T-1 (400 GPM Amine · E — 1	03001 Gas Absorber MAF-3200A (55 MMSCFD · D — 5
02003 Outlet Scrubber V-1 (400 GPM Amine · E — 2	03002 Vertical Scrubber MBF-1240A (55 MM · E — 3
02003A Gas Cooler AC-4 (400 GPM Amine Pla · E — 0 open	03003 Glycol Region (55 MMSCFD Dehy Trai · F — 0 open
02004 Amine Flash Tank V-2 (400 GPM Amin · E — 6	03004 Glycol Region — Filters & Pump · D — 2
02005 Amine Filters F-1 / F-2 (400 GPM A · E — 7	04001 Slop Tanks TK-41001...41004 · E — 6
02006 L/R Exchanger E-1 (400 GPM Amine P · E — 3	07001 Air Compressor / Receivers (SK-700 · G — 9
02007 Amine Booster Pumps / Lean Solids · E — 6	07002 JT Fuel Skid (V-100/200/300, E-400 · F — 8
02008 Reflux Pumps P-2A/B (400 GPM Amine · E — 1	07003 FG Coalescing Filter F-71001 · E — 5
02009 Amine Still / Reflux Accumulator T · F — 2	07004 Blowcase System V-71001...V-71004 · E — 7
02010 Amine Solution Cooler / Reflux Con · E — 0 open	07005 Methanol Tank TK-71001 · C — 0 open
02011 HP Amine Pumps P-3A/B (400 GPM Ami · G — 5	07006 Generator Fuel Gas (Generators I / · D — 0 open
02012 Reboiler Skid H-1 / E-2 (400 GPM A · F — 4	08001 Amine Makeup Tank TK-81003 / P-810 · E — 0 open
02013 T.O. for 400 GPM Amine Train & · F — 5	08002 TEG Glycol Makeup Tank TK-81005 / · E — 0 open
02014 BTEX for 55 MMSCFD Dehy Unit-I (V- · F — 2	08003 Chemetherm Makeup Tank TK-81007 / · E — 0 open
02115 T.O. for 640 GPM Amine Train & · F — 5	08004 Water Makeup Tank TK-81009 / TK-81 · E — 0 open
02015A Exhaust Stack / Hot Oil Heater H-2 · E — 1	08005 Anti Foam Tank TK-81011 / P-81011 · B — 3
02015B Side Stream Filter & Circulati · E — 4	01002 Gas Cooler · D — clean (Future Scope)

3 Open Punch List — by P&ID

Item keys carry over from Rev 0. Verbatim field text in *red italics*; notes in parentheses record wording changes between markup sets. Sheets with zero open items are listed in the index and Section 7 only.

HAY-PID-01001 Slug Catcher (X-11001 / X-12001)

Rev G

13 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
01001-12	Inlet PSV vent	—	Red-drawn ½" valve w/ cap labeled <i>“½”</i> on vent tailpipe below inlet PSV — install.	Piping	HIGH
01001-13	16"-GH-0001-F2AS	—	Red-drawn small-bore valve + cap on stub near 16"x10" control valve cloud — install.	Piping	HIGH
01001-16	16"-GH-0003-F2AS	—	Red-drawn valve + blind flange on stub at elbow — install.	Piping	HIGH
01001-17	16"-GH-0006-F2AS	—	Red-drawn vertical valve + cap teed up off line — install.	Piping	HIGH
01001-19	X-11001 level bridle	LIT-1001	<i>“4x2 / Remove”</i> + strike on ½" valve between LIT-1001 and control override — remove valve; verify 4"x2" reducer requirement.	I&E	HIGH
01001-20	X-12001 level bridle	LIT-1002	Same markup as 01001-19 at X-12001 — remove ½" valve; verify 4"x2" reducer.	I&E	HIGH
01001-21	X-11001 bridle	LS-1001	Red <i>“2”</i> (twice) at bridle isolation valves — verify/upsized isolation valves to 2".	Piping	MED
01001-22	X-12001 bridle	LS-1002	Red <i>“2”</i> (twice) — same verification at X-12001 bridle.	Piping	MED
01001-23	4"-LH-0002-F2AS-1½"H-ET	X-11001	Red-drawn drain valve + plug on stub — install.	Piping	HIGH
01001-24	4"-LH-0005-F2AS-1½"H-ET	X-12001	Red-drawn drain valve + plug on stub — install.	Piping	HIGH
01001-33	12"-GH-0010-F2AS	PT-1001A	Pencil note <i>“P-T 1001A”</i> with curved leader re-tapping PT-1001A onto the 12" line — relocate tap. (Red cloud/“Wire” from earlier markup removed; graphite mark remains on current set.)	I&E	HIGH
01001-37	3"-LH-0011-F2AS-1½"H-ET	—	Red <i>“2"-LH-0009-F2AS-1H-ET”</i> added above unlabeled line; flag corrected to <i>“Coalescing Filter / HAY-PID-02101”</i> ; <i>“Δ 3x2”</i> at branch — correct line tags on drawing; verify 3"x2" reducer at branch.	Engineering	MED
01001-38	2"-DR-7003-F2AS	—	Red-drawn new line + off-page flag to Liquid Blowcase (HAY-PID-07004) — add line to drawing; verify field tie-in (matches 07004 sheet markup).	Engineering	MED

HAY-PID-01003 Gas Scrubber (V-21001 / V-22001)

Rev E

7 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
01003-07	2"-GH-0027-F2AS / 10"-GH-0026-F2AS	—	Pink/red triangle on 1" vent valve at junction (to Coalescing Filter) — verify/install 1" valve.	Piping	HIGH
01003-09	V-21001 bridle	LIT-1003 / LSH-1003	<i>“Remove”</i> + strike — remove ½" valve/plug between instruments.	I&E	HIGH
01003-10	V-21001 bridle	—	Red <i>“3”</i> (twice) at bridle isolation valves — verify/upsized to 3".	Piping	MED
01003-11	V-22001 bridle	LIT-1004	<i>“Remove”</i> + strike — remove ½" valve at V-22001 bridle.	I&E	HIGH
01003-12	V-22001 bridle	—	Red <i>“3”</i> (twice) — verify/upsized bridle isolation valves to 3".	Piping	MED
01003-15	2"-LH-0010-F2AS boot drain	—	Red-drawn ½" valve + plug teed off 2" boot-drain stub — install.	Piping	HIGH
01003-23	3"-LH-0011-F2AS-1"H-ET	—	Red <i>“3”</i> size prefix added; flag sheet number struck, <i>“02101”</i> written — correct line size & continuation sheet on drawing.	Engineering	LOW

HAY-PID-02001 Inlet Coalescing Filter S-1100 (400 GPM Amine Plant)

Rev F

5 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02001-01	10"-GH-0028-F2AS	—	Red sketch retained: two tap valves w/ bleeds flanking an orifice-flange pair on the 10" inlet — install per sketch. (Earlier <i>“RO + FIT ?”</i> question text removed on current set — confirm final metering disposition, Q2.)	Engineering	HIGH
02001-07	2"-C-0201-F2AS-1.5"H-ET	LCV-1100B	<i>“Shoes”</i> — install two pipe shoes on 2" condensate lines at station.	Piping	HIGH
02001-13	Inlet riser u/s N1	—	Red <i>“½”</i> + arrowhead at coupling — install ½" valve/plug at open coupling.	Piping	HIGH

02001-14	2"-LH-0009-F2AS	—	Red <i>"2"-LH-0009-F2AS / P&ID-01003"</i> + tie of 2"-C-0201 drop into gas outlet header — update drawing; verify field tie-in. Merged: covers former 01003-22 (same continuation marked on HAY-PID-01003).	Engineering	MED
02001-16	Outlet header flag	—	Sheet number struck, <i>"01001"</i> written — correct continuation flag.	Engineering	LOW

HAY-PID-02002 Amine Contactor T-1 (400 GPM Amine Plant) Rev E

1 open item

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02002-08	Lower head, near N9	—	<i>"Remove"</i> + X — remove 2" capped valve/stub.	Piping	HIGH

HAY-PID-02003 Outlet Scrubber V-1 (400 GPM Amine Plant) Rev E

2 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02003-03	V-1 nozzle N3 (2")	PI V-1	<i>"Remove"</i> + X — remove small gauge/bleed valve at N3 between PI isolation and vessel.	I&E	HIGH
02003-08	LV V-1 bypass loop	—	Red-drawn small valve + solid plug immediately d/s of the control valve (before the downstream 1" CB203) — install bleed/drain valve + plug. NEW mark on current set.	Piping	HIGH

HAY-PID-02004 Amine Flash Tank V-2 (400 GPM Amine Plant) Rev E

6 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02004-01	8"-RA-0010-F2AS-SR	—	Red-drawn ½" valve + cap labeled <i>"½"</i> on top of rich amine run — install high-point vent.	Piping	HIGH
02004-05	V-2 nozzle M4 (4")	—	Small red check at flange pair (KMTD/MPLX break) — field verification mark; confirm intent/torque status.	Mechanical	MED
02004-06	LG V-2A bridle, N8	2" GVTONT CSO	Red <i>"CB10"</i> — change upper bridle valve spec GVTONT → CB10 (retain CSO).	Piping	HIGH
02004-07	LG V-2A bridle, N9	2" GVTONT CSO	Red <i>"CB10"</i> — change lower bridle valve to CB10.	Piping	HIGH
02004-10	1"-DR-0003-A2AS	—	<i>"Remove By MPLX"</i> + X — remove flanged drain valve adjacent saddle.	Piping	HIGH
02004-13	PCV V-2B bypass drop leg	—	Red-drawn valve + solid plug at the low-point corner of the bypass leg — install low-point drain valve + plug. NEW mark on current set.	Piping	HIGH

HAY-PID-02005 Amine Filters F-1 / F-2 (400 GPM Amine Plant) Rev E

7 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02005-07	F-1 meter run	—	Red-drawn ¾" valve + plug below FE F-1 — install vent/bleed.	Piping	HIGH
02005-08	6"-RA-0013-A2-1.5"H-SR	—	Red-drawn ¾" valve + plug near F-1 outlet riser — install drain/vent.	Piping	HIGH
02005-10	1"-V-0071-A1	PSV F-2 tail	<i>"Doesn't Exist."</i> + X on printed 1" CSO valve — valve shown on drawing is not in field.	Piping	CRITICAL
02005-11	1"-V-0071-A1	—	<i>"Needs to Be Added Inside Skid limits."</i> — install the missing 1" CSO valve + tailpipe work inside skid.	Piping	CRITICAL
02005-12	6" outlet header	—	Red-drawn valve + plug (size unmarked, ~¾") on header toward L/R exchanger — install low-point drain.	Piping	HIGH
02005-13	2"-DR-0100-A1/A2	—	Red continuation flag <i>"2-DR-0100-A2 / HAY-PID-02016"</i> at west end of drain header — correct drawing continuity on this sheet and HAY-PID-02006; verify field tie-in. Merged: covers former 02006-03 (same flag correction on the mate sheet).	Engineering	MED
02005-14	2"-DR-0100-A1 (east)	—	Red emphasis chevrons on printed drain flag — verify/connect continuation to HAY-PID-02006.	Piping	MED

HAY-PID-02006 L/R Exchanger E-1 (400 GPM Amine Plant) Rev E

3 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02006-01	6"-LA-0022-A1	TI/TW E-1D	Red cloud + tag <i>"E-1D"</i> on unlabeled TI/TW bubble — install & tag temperature indicator with thermowell on E-1 lean amine outlet.	I&E	HIGH
02006-02	6" crossover branch	—	Red-drawn valve + plug (no size) upstream of 6" block valve — install vent/drain bleed.	Piping	HIGH

02006-04	2"-DR-0100-A1 (east)	—	Red emphasis on printed flag to HAY-PID-02007 — verify continuation/tie-in.	Piping	MED
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HAY-PID-02007 Amine Booster Pumps / Lean Solids Filter (P-1A/B, F-4)

Rev E

6 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02007-01	6"-LA-0018-A1-1.5"H	P-1A discharge	Red <i>"½"</i> + drawn valve between check and block — install ½" valve (vent/gauge).	Piping	HIGH
02007-02	6"-LA-0019-A1-1.5"H	P-1B discharge	Red <i>"½"</i> + drawn valve — install ½" valve.	Piping	HIGH
02007-03	1"-V-0076-A1	1" C1203F LSD	<i>"Doesn't Exist."</i> [sic] + cloud — printed 1" shutdown valve not in field; reconcile install vs drawing revision. (Eng. Question Q13)	Engineering	HIGH
02007-05	6"-LA-0021-A1-1.5"H	6" CV10	<i>"Doesn't Exist."</i> + cloud — printed 6" valve on lean amine outlet not in field; reconcile. (Eng. Question Q13)	Engineering	HIGH
02007-06	2"-DR-0100-A1 (east)	—	<i>"Removed"</i> + cloud + cut marks — drain header terminus piping removed in field; update drawing / confirm extent.	Engineering	MED
02007-08	Spare-filter bridges	PDI F-4B / F-4C	Red tick marks on both PDI balloons — verify gauges installed/functional.	I&E	MED

HAY-PID-02008 Reflux Pumps P-2A/B (400 GPM Amine Plant)

Rev E

1 open item

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02008-04	2"-RF-0035-A2-1.5"H-ET-SR	LV T-2	Red <i>"¾"</i> + drawn valve below line — install ¾" drain/bleed between block and check.	Piping	HIGH

HAY-PID-02009 Amine Still / Reflux Accumulator T-2

Rev F

2 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02009-02	10"-LA-0017-A2-1.5"H	—	Red <i>"1"</i> ×3 + drawn valves — install two 1" high-point vents and one 1" drain on lean amine to reboiler.	Piping	HIGH
02009-05	Nozzle C2 gauges	PI T-2B, TI T-2B	Red instrument range edits at nozzle C2: <i>"0-100#"</i> over PI T-2B and <i>"50-300F"</i> over TI T-2B — verify/replace gauges to marked ranges (current set reads 50-300F).	I&E	MED

HAY-PID-02011 HP Amine Pumps P-3A/B (400 GPM Amine Plant)

Rev G

5 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02011-05	6"-LA-0025-G2	PIT P-3A2	Red bubble + two drawn root valves at new tap between CV90 and check — relocate PIT P-3A2 tap here; install two small-bore root valves.	I&E	HIGH
02011-08	6"-LA-0024-G2	PI/PIT P-3D	Red cloud around PI/PIT P-3D + arc arrows to the new tap on 6"-LA-0024-G2 — relocate tap (mirror of 02011-05). ("Wire" call-out removed on current set — wiring scope cleared.)	I&E	HIGH
02011-14	6"-LA-0025-F2	FIT/FIC P-3	<i>"Missing"</i> + pink highlight — install FIT P-3 dP transmitter, both impulse manifold valve sets, and tubing on common discharge meter.	I&E	CRITICAL
02011-16	P-3A suction drop	6" cone strainer	Red doodle at temporary strainer — verify installed; remove after flushing per commissioning plan.	Piping	HIGH
02011-17	P-3B suction drop	6" cone strainer	Same verification at P-3B.	Piping	HIGH

HAY-PID-02012 Reboiler Skid H-1 / E-2 (400 GPM Amine Plant)

Rev F

4 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02012-01	E-2 level bridles	LG/LT E-2A, E-2B	Large cloud: <i>"Change all of Them to Gate Valves"</i> — replace all bridle isolation valves (½", ¾" CB203; 2" CB10 CSO) with gate valves; count in field before ordering.	Piping	HIGH
02012-07	24"-LA-0029-A2-1.5"H	—	Red-drawn 1" stub/ø glyph on 24"-LA-0029-A2-1.5"H adjacent TE/TW E-2E — plug open 1" connection. ("Needs plug" text removed; drawn mark remains.)	Piping	CRITICAL
02012-10	E-2 shell (S4-S1)	—	Red-drawn 2" valve at shell connection — install (field addition).	Piping	HIGH

02012-11	TCV E-2 station	—	Red “6” at four station valves — verify actual valve sizes vs drawing; revise drawing or valves. (Eng. Question Q9)	Engineering	MED
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HAY-PID-02013 T.O. for 400 GPM Amine Train & Dehy Train I (VCU-101)

Rev F

5 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02013-01	2"-GH-7004-A2	—	Red-drawn 2" valve u/s of PCV-100 — install 2" fuel gas isolation valve.	Piping	HIGH
02013-02	2"-FG-2220-A2	FE/FIT-100	Red-drawn orifice marks + “2” flanking the FIT-100 tap on 2"-FG-2220-A2 — install FE-100 restriction orifice + FIT-100; terminate previously pulled wire. (Pink/“Missing... wire pulled” text removed; drawn marks remain.)	I&E	CRITICAL
02013-05	14"-AG-0033-A2AS-SR		Red-drawn high-point vent (valve + plug) before KOD-301 — install (likely 1"; verify size).	Piping	HIGH
02013-07	KOD-301 top	—	Red-drawn valve at PSV/inlet nozzle connection — install small-bore valve.	Piping	HIGH
02013-08	KOD-301 drain	—	Red-drawn 2" valve at drain termination d/s of flame arrestor — install.	Piping	HIGH

HAY-PID-02014 BTEX for 55 MMSCFD Dehy Unit-I (V-7000A, HZZ-1100A, TK-1200A)

Rev F

2 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02014-03	D — system by-pass vent	—	Cloud: “Missing / Not on ISOs / SKID LIMITS” — fabricate/install 2" NPT bypass-vent piping; add to ISOs. (Eng. Question Q6)	Piping	HIGH
02014-10	2" NPT bypass-vent run	—	“Missing” + cloud/pink — fabricate/install segment from D vent past HZZ-1100A to BTEX outlet drop near SP-1100A.	Piping	HIGH

HAY-PID-02115 T.O. for 640 GPM Amine Train & Dehy Train II (VCU-102)

Rev F

5 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02115-01	2"-GH-7005-A2	—	Red-drawn 2" valve u/s of PCV-103 — install 2" fuel gas isolation valve.	Piping	HIGH
02115-02	2"-FG-2222-A2	FE/FIT-101	Red-drawn orifice marks flanking FE/FIT-101 tap on 2"-FG-2222-A2 — install FE-101 orifice + FIT-101; terminate pulled wire. (Text removed; drawn marks remain.)	I&E	CRITICAL
02115-05	14"-AG-0040-A2AS-SR		Red-drawn high-point vent (valve + plug) before KOD-302 — install (verify size).	Piping	HIGH
02115-07	KOD-302 14" inlet drop	—	Red-drawn valve at inlet connection — install (size per connection; verify).	Piping	HIGH
02115-09	KOD-302 2" drain	—	Red-drawn valve + termination at open end — install 2" valve (+ flame arrestor per KOD-301 pattern).	Piping	HIGH

HAY-PID-02015A Exhaust Stack / Hot Oil Heater H-27001A (Hot Oil Unit-I)

Rev E

1 open item

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02015A-03	Heater outlet detail	PI-27002A	Cloud: “Missing / Not on ISO’s” — outlet pressure-gauge assembly (0-160 psig, ½" NPT valve, siphon; shipped loose per Note 8, MPLX installs) missing. Merged: includes the ½" tap required on 8"-HM-187-C2-1.5"H (former 02015A-04).	I&E	HIGH

HAY-PID-02015B Side Stream Filter & Circulation Pumps (Hot Oil Unit-I)

Rev E

4 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02015B-02	6"-40-TF-106 → filter	SSI-30001A	Red-drawn routing from the 1" branch (GL4SW3) to the SSI-30001A filter inlet — install side-stream supply per markup. (Size labels removed on current set — verify 1" in field.)	Piping	HIGH
02015B-03	Filter hookup		Red-drawn loop through filter inlet/outlet (1" 3000# SW nozzles) — re-route/complete filter piping per sketch.	Piping	HIGH
02015B-04	SSI-300017 riser	—	Red trace at header crossing — crossing only (no connection); as-built clarification.	Engineering	LOW
02015B-05	6"-40-TF-110	—	Red-drawn vertical drop tying into 6"-40-TF-110 — install side-stream return tie-in per markup. (Size label removed — verify 1" in field.)	Piping	HIGH

HAY-PID-02015C Expansion Tank TK-45001A (Hot Oil Unit-I)

Rev E

6 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
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02015C-01	¼"-GH-2001-A2	PCV-2001	Red hatch marks on ¼"-GH-2001-A2 at the PCV-2001 station inlet — blanket-gas regulator station (set 30 psig) still flagged not installed. (Earlier “¾" valve installed on adjacent 3" FG pipe” deviation note removed — reconcile with engineering, Q5.)	I&E	CRITICAL
02015C-02	To nozzle C3	—	Red hatch marks along the ¼" blanket-gas run to nozzle C3 — install ¼" tubing/branch, full run across HAY-PID-02015C and 02015D. Merged: covers former 02015D-01 (matching hatch on the blower sheet).	I&E	CRITICAL
02015C-03	Nozzles C1, C3	—	Red checks on both 150# RF flanges — complete/verify hookups (with -02).	Piping	HIGH
02015C-06	Nozzle C7	TP-E6	Cloud: <i>“Where to?” / “Don't Exist - Not on ISO's”</i> — test point connection absent and unrouted; engineering to clarify. (Eng. Question Q4)	Engineering	HIGH
02015C-07	8"-HM-187-C2-1.5"H	—	Cloud at the 8" gooseneck on 8"-HM-187-C2-1.5"H , text now <i>“Doesn't Exist - not on ISO's”</i> — install loop/seal assembly; add to ISOs (Q4).	Piping	HIGH
02015C-08	8"-HM-180 / 8"-HM-187	FCV-45001A	Cloud + pink: <i>“Doesn't Exist - Not on ISO's” / “Missing”</i> — 8" control-valve station absent entirely; engineering to confirm requirement, then procure/install. (Eng. Question Q4)	Engineering	CRITICAL

HAY-PID-02015D
Combustion Air Blower / Burner BRN-20001A (Hot Oil Unit-I)

Rev E

2 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02015D-02	1½"-80-NG-204	XV-20003A	Pink highlight — burner main-gas vent valve assembly (1½" XV, FO orifice, vent gooseneck) missing; install (vent piping by MPLX, Note 2).	I&E	CRITICAL
02015D-03	¾"-80-NG-205	XV-20006A	<i>“Missing”</i> (two-headed arrow, also to -02) — pilot-gas vent valve assembly (¾" XV w/ solenoid, FO orifice, gooseneck) missing; install.	I&E	CRITICAL

HAY-PID-02016
Catch Tank TK-2001 / TK-2002 (400 GPM Amine Plant)

Rev D

2 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
02016-03	Drop to N4 (½")	—	Red fitting stack <i>“1x½”</i> — install union/flange pair, valve, 1"x½" swage at N4.	Piping	HIGH
02016-13	TK-2002 header drops	—	Red routing clarifications (break squiggles at 1½"-PV-0002/0003-A2 × 1"-DA-0001-A2 / 1"-V-0094-A2 crossings) — verify as-built tie-in locations; update drawing.	Engineering	MED

HAY-PID-03001
Gas Absorber MAF-3200A (55 MMSCFD Dehy Train-I)

Rev D

5 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
03001-03	PSV mount	PSV-3200A	Red <i>“1½”</i> + drawn nozzle at vessel wall — verify PSV connection nozzle size 1½" vs printed 1"x1". (Eng. Question Q12)	Engineering	HIGH
03001-05	RO-3201A flanges	—	<i>“Bleed Rings”</i> — install bleed ring(s) with bleed valves at orifice flanges.	Piping	HIGH
03001-10	Nozzles E (1" liq outlet), D (1½"-RG-3001-F2)	—	Red hash ticks at vendor termination points (Note 3: piping beyond termination not provided) — complete downstream tie-ins.	Piping	HIGH
03001-11	Nozzles C (lean glycol), H (1"-GH-3012-F2)	—	Red hash ticks at vendor termination points — complete downstream tie-ins.	Piping	HIGH
03001-12	8"-GH-3004-F2	—	Red-drawn 1" plugged valve assembly (stem, valve, flange bars, plug) — install 1" valved+plugged branch (vent/sample) on dry gas line.	Piping	HIGH

HAY-PID-03002
Vertical Scrubber MBF-1240A (55 MMSCFD Dehy Train-I)

Rev E

3 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.
03002-02	8"-GH-3005-F2	—	Red-drawn 1" valve on stub at elbow above nozzle B — install 1" high-point vent.	Piping	HIGH
03002-05	FV-1240A manifold	—	Red marks at two ½" bypass/vent valves — install ½" plugs/caps (or valves); confirm intent in field.	Piping	MED
03002-19	16" riser / 2"-DA-3003-F2-1.5"H-ET	—	<i>“U/G”</i> breaks + red trace to PCV-3001 — record underground transition; verify slop-header tie-in.	Engineering	MED

HAY-PID-03004
Glycol Region — Filters & Pumps (55 MMSCFD Dehy Train-I)

Rev D

2 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT & ACTION REQUIRED</i>	DISC.	PRI.

03004-01	Flash-gas line → 1"-V-0046-A2	1" H5S	Cloud + <i>"Missing Checkvalve"</i> — install 1" H5S check valve on the flash-gas line to 1"-V-0046-A2 .	Piping	HIGH
03004-06	1½" A00-GR-04	1½" W2ZU	<i>"valve in wrong spot"</i> / <i>"Has to move"</i> — relocate valve from MAJ-1200A outlet nozzle to marked position on vertical branch.	Piping	HIGH

HAY-PID-04001 Slop Tanks TK-41001...41004

Rev E

6 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
04001-05	PVRV risers, C3 (×4)	Enardo ES-660-HF	<i>"6X4"</i> ×4 — install 6"x4" transition/reducer at each tank's PVRV riser.	Piping	HIGH
04001-06	Tank bottoms (×4)	—	Red-drawn 1" valve at each tank near manway — install four 1" drain/water-draw valves.	Piping	HIGH
04001-07	C4 nozzles (×4)	—	Red triangle at each 4" side nozzle — verify blinded; install 4" blind flanges if open.	Piping	HIGH
04001-08	2"-DA-2101-A2-1.5"H-ET	—	Red-drawn new 2" HT drain line + 2" valve + capped riser at containment (P&ID ref blank) — install; engineering to assign tie-in/sheet ref.	Piping	HIGH
04001-09	2"-DA-2[6]81-A2-1.5"H-ET [digits unclear]	—	Second red-drawn 2" HT drain line + 2" valve, tie-in "HAY-P&ID 02013" — install; verify line number with engineering.	Piping	HIGH
04001-10	Drain headers	—	Red tie-in routing arrows into 2"-DA-3003-A2 / 4"-DA-3023-A2 / 2"-DA-2051-A2 junctions — verify as-built tie-in points; update drawing.	Engineering	MED

HAY-PID-07001 Air Compressor / Receivers (SK-70001, VS-70001/2/4, TK-70001)

Rev G

9 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
07001-01	VS-70004 wet receiver	PSV-7005	<i>"Remove"</i> + cloud + X — remove PSV-7005 manifold (set 155 psig) and adjoining PI-7004/½" valve. Engineering to confirm relief philosophy change. (Eng. Question Q16)	Engineering	CRITICAL
07001-02	VS-70004 top, 6" nozzle	—	<i>"VENT"</i> + drawn gooseneck along 1½"-IA-7036-L2 — install atmospheric vent in lieu of removed PSV.	Piping	CRITICAL
07001-03	½"-IA-7040-L2	—	Red X-hatch — delete/remove line + ½" SS tube run below VS-70004.	Piping	HIGH
07001-05	VS-70001 dry receiver	PSV-7003	<i>"Remove"</i> + cloud + X — remove PSV assembly (set 155 psig). (Eng. Question Q16)	Engineering	CRITICAL
07001-06	VS-70001 top nozzle	—	<i>"Vent"</i> + <i>"← PI"</i> — install gooseneck vent + pressure gauge at top (1½"-IA-7032-L2).	Piping	HIGH
07001-07	VS-70001 shell	PI-7001	Red cloud — relocate PI-7001 to top nozzle (with -06).	I&E	HIGH
07001-09	VS-70002 dry receiver	PSV-7001	<i>"Remove"</i> + cloud + X — remove PSV assembly. (Eng. Question Q16)	Engineering	CRITICAL
07001-10	VS-70002 top nozzle	—	<i>"Vent"</i> + <i>"PI"</i> — install vent + PI (1½"-IA-7035-L2).	Piping	HIGH
07001-11	VS-70002 shell	PI-7002	Red cloud — relocate to top (with -10).	I&E	HIGH

HAY-PID-07002 JT Fuel Skid (V-100/200/300, E-400A/B, P-500A-D)

Rev F

8 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
07002-10	V-100 overhead	PCV-100A, PI-101A	Cloud + <i>"Missing"</i> — install instrument-gas regulator set: PCV-100A, PI-101A (0-150#), ¼" valve, tubing, TO SG-7 takeoff.	I&E	HIGH
07002-11	Above LCV-100	—	Red <i>"¼"</i> + drawn valve — install ¼" valve upstream of LCV-100 (size ¼" vs ¾" to be verified).	Piping	HIGH
07002-12	Below LCV-100	—	Red <i>"¼"</i> + drawn valve above 1" DSS spec break — install ¼" bleed.	Piping	HIGH
07002-13	D/s LCV-200	—	Red <i>"¼"</i> + drawn valve at 2" DSS break — install ¼" valve.	Piping	HIGH
07002-14	D/s LCV-300	—	Red <i>"¾"</i> + drawn valve at 2" DSS break — install ¾" valve (size to be verified).	Piping	HIGH
07002-15	V-300 overhead 2"-PG-300-03	Methanol inj. pt	Cloud: <i>"not connected inside skid."</i> — complete ¼" methanol-injection tie-in from pump 4.	Piping	HIGH

07002-16	3"-PG-104-A (E-400A N4)	—	Red highlighter trace, no text — segment flagged; verify installation/routing.	Piping	MED
07002-17	3"-PG-104-B (E-400B N4)	—	Red highlighter trace — verify installation/routing.	Piping	MED

HAY-PID-07003 FG Coalescing Filter F-71001

Rev E

5 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
07003-05	DP installation	PDI-7101	Cloud: <i>“By Valley.” / “Missing”</i> — install PDI-7101 assembly (manifold, ½" valves, tubing, 2"x½" taps); vendor “Valley” scope.	Vendor	HIGH
07003-06	Gauge standpipe (N5)	—	Red-drawn vent valve at top — install.	Piping	HIGH
07003-07	Stub below N6	—	Red-drawn valve + plug — install drain valve + plug above 2"-LH-7002-A2-1.5"H-ET.	Piping	HIGH
07003-08	Nozzle N8 stub	—	Red valve symbol (~¾", partially legible) at KMTD/MPLX boundary — install valve; verify size.	Piping	HIGH
07003-09	Stub below N7	—	Red-drawn valve + plug next to RO-7102 — install drain valve + plug.	Piping	HIGH

HAY-PID-07004 Blowcase System V-71001...V-71004

Rev E

7 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
07004-01	V-71001 bottom outlet	1" valve	<i>“Remove”</i> + X — remove 1" valve (into 2" drain manifold).	Piping	HIGH
07004-02	V-71002 bottom outlet	1" valve	<i>“Remove”</i> + X — remove 1" valve (to 2"-DR-7002-F2AS).	Piping	HIGH
07004-07	2"-DR-7002-F2AS	—	Red routing trace from 2"-DR-7002-F2AS into the bottom drain header — complete routing per markup. (Earlier “2"/Add” valve portion removed on current set — valve add cleared.)	Piping	HIGH
07004-10	V-71003 sensing line	PC/PV-71003C	Red-drawn valve + <i>“¼”</i> — install ¼" valve in sensing line.	I&E	HIGH
07004-11	LG-71003 gauge drain	—	Red tick + triangle at 1"x¾" tap — install ¾" bull plug / verify drain valve.	Piping	HIGH
07004-12	LG-71004 gauge drain	—	Red marks below lower gauge valve — install ¾" drain valve + plug.	Piping	HIGH
07004-13	V-71003/71004 outfall	Open drain	Cloud around OPEN DRAIN at the V-71003/V-71004 outfall, text now <i>“Where to / No ISO”</i> — drain destination undefined and no isometric exists; engineering to route (Q19), then tie in.	Engineering	HIGH

HAY-PID-08005 Anti Foam Tank TK-81011 / P-81011

Rev B

3 open items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
08005-04	End of discharge run	—	Red-drawn valve + <i>“2”</i> (+ “inst...” partially legible) — install 2" isolation valve upstream of anti-foam distribution split.	Piping	HIGH
08005-05	Distribution split → P-1A/B	2" ball valve	<i>“Missing.”</i> + cloud — install 2" ball valve on through-run to HAY-PID-02007.	Piping	HIGH
08005-06	Distribution area	—	<i>“Need to Correct Iso To Match P&ID”</i> — revise piping isometric to match P&ID.	Engineering	LOW

4 Material Takeoff (BOM) — Open Items Only

Rebuilt from the 137 open items. Unknown ratings/materials read “Verify in Piping Material Specification” (V-PMS). Rev 0 BOM lines for closed items are withdrawn.

QTY	DESCRIPTION	SIZE	RATING	MATERIAL	SPEC	TAG	LINE NUMBER	NOTES / PUNCH REF
2	Flow element (restriction orifice) + FIT transmitter; terminate pulled wire	2" line	V-PMS	V-PMS	A2	FE/FIT-100, FE/FIT-101	2"-FG-2220-A2 / 2"-FG-2222-A2	02013-02 / 02115-02

1	dP flow transmitter + 2 impulse manifold valve sets + tubing	—	V-PMS	—	F2	FIT P-3	6"-LA-0025-F2	02011-14
1	PDI dP assembly: manifold, ½" valves, tubing, 2"x½" taps ("By Valley" cloud)	½"	V-PMS	V-PMS	—	PDI-7101	F-71001	07003-05
1	Regulator set: PCV + PI 0-150# + ¼" valve + tubing	¼"	V-PMS	V-PMS	—	PCV-100A / PI-101A	V-100 overhead	07002-10
1	Blanket-gas regulator + ¼" fittings	¼"	Set 30 PSIG	V-PMS	A2	PCV-2001	¾" -GH-2001-A2	02015C-01 (Q5)
lot	¼" blanket-gas tubing/branch, full run across 02015C + 02015D	¼"	—	SS	—	—	¾" -GH-2001-A2	02015C-02 (merged)
1	Burner main-gas vent valve assembly: XV + FO orifice + vent gooseneck	1½"	V-PMS	V-PMS	NG	XV-20003A	1½" -80-NG-204	02015D-02
1	Burner pilot-gas vent valve assembly: XV w/ solenoid + FO orifice + gooseneck	¾"	V-PMS	V-PMS	NG	XV-20006A	¾" -80-NG-205	02015D-03
1	Control valve station w/ actuator — HOLD pending Q4 confirmation	8"	V-PMS	V-PMS	C2	FCV-45001A	8"-HM-180 / -187	02015C-08 (long lead)
1	Pipe loop/seal (gooseneck) assembly w/ plug/blind — HOLD pending Q4	8"	V-PMS	V-PMS	C2	—	8"-HM-187-C2-1.5"H	02015C-07
1	Pressure gauge assembly 0-160 PSIG w/ ½" NPT valve + siphon (shipped loose — locate) + ½" threadolet tap on 8" line	½"	V-PMS	V-PMS	C2	PI-27002A	8"-HM-187-C2-1.5"H	02015A-03 (merged)
1	TI + thermowell	¾"	V-PMS	V-PMS	A1	TI/TW E-1D	6"-LA-0022-A1	02006-01
2	Pressure gauge (receiver top mount, with vent goosenecks x3)	½"	V-PMS	V-PMS	L2	VS-70001 / VS-70002 / VS-70004	—	07001-02/-06/-10 (Q16)
1	Check valve	1" (H5S)	V-PMS	V-PMS	A00	—	flash gas → 1"-V-0046-A2	03004-01
1	Valve, CSO + tailpipe completion inside skid limits	1"	V-PMS	V-PMS	A1	—	1"-V-0071-A1	02005-10/-11
1+	Bleed ring set w/ bleed valves at orifice flanges	1" 6000#	V-PMS	V-PMS	F2	R0-3201A	MAF-3200A glycol drain	03001-05
1	Valved + plugged branch assembly (o-let, valve, plug)	1"	6000# taps	V-PMS	F2	—	8"-GH-3004-F2	03001-12
1	Orifice-flange pair + 2 tap valves w/ bleeds (per field sketch)	10" line	V-PMS	V-PMS	F2AS	—	10"-GH-0026-F2AS	02001-01 (Q2)

±9	Valves, small-bore w/ caps or plugs (vents/bleeds/drains)	½"	V-PMS	V-PMS	per line class	—	various	01001-12, 01003-15, 02001-13, 02004-01, 02007-01/-02, 03002-05 (plugs), 02003-08*, 02004-13*
±7	Valves w/ plugs (drains/vents)	¾"	V-PMS	V-PMS	per line class	—	various	02005-07/-08, 02008-04, 07003-07/-09, 07004-11 (plug)/-12
±10	Valves (incl. 3 on 10"-LA-0017, 4 slop-tank drains, high-point vent on 8"-GH-3005)	1"	V-PMS	V-PMS	per line class	—	various	02009-02, 03002-02, 04001-06, 02012-07 (plug only)
±4	Valves, ¼" (LCV stations, sensing line)	¾"	V-PMS	V-PMS	DSS / per class	—	JT skid, blowcase	07002-11/-12/-13, 07004-10 (07002-14 is ¾")
±7	Valves (fuel-gas isolations x2, KOD drains x2, E-2 shell, anti-foam x2)	2"	V-PMS	V-PMS	per line class	—	various	02012-10, 02013-01/-08, 02115-01/-09, 08005-04/-05
2	Valve spec change GVTONT → CB10 (CSO)	2"	V-PMS	V-PMS	CB10	LG V-2A bridle	V-2 N8/N9	02004-06/-07
Field count	Gate valves to replace E-2 bridle valves (½", ¾" CB203; 2" CB10 CSO)	½"-2"	V-PMS	V-PMS	—	LG/LT E-2A/B	E-2 bridles	02012-01
2	High-point vent assemblies (valve + plug; size unmarked, likely 1")	TBD	V-PMS	V-PMS	A2AS	—	14"-AG-0033 / 14"-AG-0040	02013-05 / 02115-05
2	Valves at KOD-301/302 top connections (size per connection)	TBD	V-PMS	V-PMS	—	KOD-301 / KOD-302	—	02013-07 / 02115-07
2	Root valves + fittings for relocated PIT taps	½"	V-PMS	V-PMS	G2	PIT P-3A2 / PIT P-3D	6"-LA-0025/0024-G2	02011-05/-08
1	Valve + blind flange on stub	SB	V-PMS	V-PMS	F2AS	—	16"-GH-0003-F2AS	01001-16 (+ caps: -13/-17)
±2	Reducers 4"x2" (bridle LIT mounts — verify)	4"x2"	V-PMS	V-PMS	—	LIT-1001/-1002	X-11001/X-12001 bridles	01001-19/-20; 3"x2" verify 01001-37
4	Reducer/transition, PVRV riser	6"x4"	V-PMS	V-PMS	—	Enardo risers	TK-41001...41004 C3	04001-05
±4	Blind flange (verify nozzles open) + gaskets/bolting	4"	V-PMS	V-PMS	—	Slop tank C4 nozzles	TK-41001...41004	04001-07
1	Swage + union/flange pair + valve	1"x½"	V-PMS	V-PMS	F2AS	TK-2001 N4 drop	½"-C-7001-F2AS	02016-03
2	New heat-traced drain lines: 2" pipe + fittings + 2" valves + heat trace/insulation	2"	V-PMS	V-PMS	A2, 1.5"H-ET	—	2"-DA-2101 + second line (Q18)	04001-08/-09/-10
lot	Side-stream filter loop piping + branch fittings (size labels removed — verify 1")	1"	V-PMS	V-PMS	—	SSI-30001A	6"-40-TF-106 / -110	02015B-02...-05
lot	2" NPT bypass-vent piping (BTEx), per markup — add to ISOs	2" NPT	V-PMS	V-PMS	—	—	D vent → C outlet	02014-03/-10 (Q6)
1	Pipe shoe	2"	—	CS	—	—	2"-C-0201-F2AS-1.5"H-ET	02001-07

2	Replacement gauges if ranges wrong: PI 0-100#, TI 50-300F	—	—	—	—	PI T-2B / TI T-2B	T-2 nozzle C2	02009-05
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* new items. Withdrawn from Rev 0 BOM (marks closed): all PSVs and tailpipes, 14"/3" VF valves, KOD level-switch stacks and gauges, orifice plates w/ specified bores, X-2 static mixer, cone strainers/FDs, U-bolts/shoes/supports (except one shoe), platforms/grating, grout, grounding, instrument cable, electrical covers, Garlock gasket, truck connections, enviro box.

5 Engineering Questions — Open

- Q2 — Coalescing filter inlet (HAY-PID-02001):** the orifice/tap sketch remains but the “RO + FIT ?” question text was removed — confirm final metering disposition so 02001-01 can be executed. (Companion Q1 on the slug catcher was fully removed — see Section 7.)
- Q4 — Hot Oil expansion tank (HAY-PID-02015C):** FCV-45001A station, the 8" gooseneck, and test point TP-E6 are all clouded “*Doesn’t Exist - not on ISO’s*”. Confirm scope, issue ISOs, and release the 8" control valve for procurement if required (long lead). (02015C-06/-07/-08)
- Q5 — Blanket gas (HAY-PID-02015C):** PCV-2001 station still hatched as not installed; the earlier field-deviation note (“¾" valve on adjacent 3" FG pipe”) was removed. Install per design or formalize the deviation. (02015C-01)
- Q6 — BTEX bypass vent (HAY-PID-02014):** 2" NPT vent piping still clouded “Missing / Not on ISOs / Skid limits” — issue ISO/routing. (02014-03/-10)
- Q9 — TCV E-2 station sizes (HAY-PID-02012):** red “6” marks remain at four station valves — verify installed sizes vs drawing. (02012-11)
- Q12 — PSV-3200A nozzle (HAY-PID-03001):** red 1½" vs printed 1"x1" NPT still marked — confirm connection size. (03001-03)
- Q13 — “Doesn’t exist” valves (HAY-PID-02007 / 02005):** printed 1" C1203F LSD valve, 6" CV10 lean-amine valve, and the PSV F-2 tailpipe 1" CSO remain marked absent — install per drawing or revise drawings. (02007-03/-05, 02005-10/-11)
- Q16 — Air-system relief redesign (HAY-PID-07001):** the PSV-7001/-7003/-7005 removals with replacement vents + relocated PIs remain fully marked. Confirm the redesign is engineered and code-compliant BEFORE removing relief devices. (07001-01...-11)
- Q18 — Slop-tank drain lines (HAY-PID-04001):** both red-drawn 2" lines remain; one tie-in reference still blank, the other line number partially legible — assign line numbers/sheet refs. (04001-08/-09/-10)
- Q19 (new) — Blowcase open drain (HAY-PID-07004):** cloud now reads “Where to / No ISO” — define the drain destination and issue the ISO, then tie in. (07004-13)

Retired with their marks (see Section 7): Q1 (slug-catcher metering), Q3 (relief to 165# vessel — safety-flagged), Q7 (pump asterisks), Q8 (Sigma torque scope), Q10 (CSC/CSO), Q11 (½" condensate upsize), Q14 (LSH hookup), Q15 (platform), Q17 (size-hold releases tied to closed PSV items).

6 Procurement List

PRIORITY	ORDER	QTY	NOTES
CRITICAL	FE/FIT-100 & -101 orifices + transmitters; FIT P-3 + manifolds	3 sets	wire already pulled at VCU meters
CRITICAL	Blanket-gas: PCV-2001 + ¼" tubing/fittings; burner vent assemblies XV-20003A / XV-20006A	1+lot+2	hot-oil startup blockers
CRITICAL	FCV-45001A 8" CV station; 8" gooseneck assembly	1+1	HOLD Q4 — release scope first; long lead
HIGH	Valve package: ¼"x4, ½"x9±, ¾"x7±, 1"x10±, 2"x7±, 2" CB10 ×2, 1" H5S check, 1" CSO, KOD-top valves ×2, HP-vent assemblies ×2, gate-valve swap (field count)	~48	order to line-class specs — V-PMS
HIGH	PDI-7101 assembly; PCV-100A/PI-101A set; PI-27002A gauge assy + ½" threadolet; TI/TW E-1D; receiver PIs ×2; bleed rings; orifice-flange pair + tap valves (Q2)	—	PI-27002A is shipped-loose — locate first
HIGH	Piping lots: 2" HT slop drains ×2 (+heat trace), 1" side-stream loop, 2" NPT BTEX vent, reducers (6"x4" ×4, 4"x2" ±2, 1"x½"), 4" blinds ±4, 1 shoe	lot	Q18/Q6 refs to be assigned
MED	Replacement gauges PI 0-100# / TI 50-300F (if range verification fails)	2	02009-05

7 Disposition Register — Items Removed in this Correction

All 430 items from the Rev 0 review were individually re-verified against the current redline set. Items below are **closed from the open punch list** — either their redline mark no longer appears on the current authoritative set (reason “not on current set”), or they duplicated another item (reason “duplicate”). Item keys are never reused.

Merged duplicates (3)

RETIRED KEY	MERGED INTO	SCOPE
01003-22	02001-14	2"-LH-0009-F2AS continuation/tie between Gas Scrubber and Coalescing Filter — one field action marked on both sheets.
02006-03	02005-13	Drain header 2-DR-0100-A2 continuation-flag correction — same correction marked on both mate sheets.
02015D-01	02015C-02	Fuel-gas blanket ¼" branch + tubing — one continuous installation marked across both hot-oil sheets.

Safety review before closing (marks removed from the current set — recommend field confirmation, not silent closure)

- **07004-08 — “Installed Backwards” 2" check valve** on 2"-DR-7001-F2AS : the note was deleted from the markup. Verify correct orientation in the field before treating as resolved.
- **02011-02 — PSV P-3C (1440 psig) relieving toward a 165# MAWP vessel** (former Q3): question note deleted. Confirm engineering formally dispositioned the relief path.
- **Missing-PSV marks removed:** V-22001 package (01003-06), PSV-1100 (02001-03), PSV E-2 (02012-03), PSV AC-1 (02010-01), KOD PSVs & level switches (02013-06/-10, 02115-06/-12), PSV-7002 station (07001-14), PSV-7006 (07006-01), PSV-7101 tailpipe (07003-01), T-1 vent tailpipe (02002-02). These were relief-critical. Confirm each device is physically installed at PSSR walkdown before startup.
- **01001-05/-06 — spectacle blinds “Roll after PSSR”:** marks removed; carry blind status in the PSSR punch list.

Closed — not on current redline set (289)

P&ID	ITEM KEYS CLOSED
HAY-PID-01001	01001-: 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 11, 14, 15, 18, 25, 26, 27, 28, 29, 30, 31, 32, 34, 35, 36, 39, 40, 41,
HAY-PID-01003	01003-: 01, 02, 03, 04, 05, 06, 08, 13, 14, 16, 17, 18, 19, 20, 21
HAY-PID-02001	02001-: 02, 03, 04, 05, 06, 08, 09, 10, 11, 12, 15
HAY-PID-02002	02002-: 01, 02, 03, 04, 05, 06, 07, 09, 10
HAY-PID-02003	02003-: 01, 02, 04, 05, 06, 07
HAY-PID-02003A	02003A-: 01
HAY-PID-02004	02004-: 02, 03, 04, 08, 09, 11, 12
HAY-PID-02005	02005-: 01, 02, 03, 04, 05, 06, 09
HAY-PID-02007	02007-: 04, 07
HAY-PID-02008	02008-: 01, 02, 03, 05
HAY-PID-02009	02009-: 01, 03, 04, 06, 07, 08, 09, 10, 11, 12
HAY-PID-02010	02010-: 01, 02, 03, 04, 05
HAY-PID-02011	02011-: 01, 02, 03, 04, 06, 07, 09, 10, 11, 12, 13, 15
HAY-PID-02012	02012-: 02, 03, 04, 05, 06, 08, 09, 12
HAY-PID-02013	02013-: 03, 04, 06, 09, 10, 11, 12, 13, 14
HAY-PID-02014	02014-: 01, 02, 04, 05, 06, 07, 08, 09, 11, 12, 13, 14
HAY-PID-02115	02115-: 03, 04, 06, 08, 10, 11, 12, 13, 14, 15, 16, 17, 18
HAY-PID-02015A	02015A-: 01, 02, 04, 05, 06, 07, 08, 09, 10, 11, 12
HAY-PID-02015B	02015B-: 01
HAY-PID-02015C	02015C-: 04, 05
HAY-PID-02016	02016-: 01, 02, 04, 05, 06, 07, 08, 09, 10, 11, 12
HAY-PID-03001	03001-: 01, 02, 04, 06, 07, 08, 09, 13, 14, 15
HAY-PID-03002	03002-: 01, 03, 04, 06, 07, 08, 09, 10, 11, 12, 13, 14, 15, 16, 17, 18
HAY-PID-03003	03003-: 01, 02, 03
HAY-PID-03004	03004-: 02, 03, 04, 05, 07, 08, 09
HAY-PID-04001	04001-: 01, 02, 03, 04, 11, 12, 13, 14, 15
HAY-PID-07001	07001-: 04, 08, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22
HAY-PID-07002	07002-: 01, 02, 03, 04, 05, 06, 07, 08, 09
HAY-PID-07003	07003-: 01, 02, 03, 04, 10, 11, 12, 13
HAY-PID-07004	07004-: 03, 04, 05, 06, 08, 09, 14, 15, 16
HAY-PID-07005	07005-: 01, 02, 03
HAY-PID-07006	07006-: 01, 02, 03, 04, 05, 06, 07, 08
HAY-PID-08001	08001-: 01
HAY-PID-08002	08002-: 01
HAY-PID-08003	08003-: 01

HAY-PID-08004	08004-: 01, 02, 03, 04, 05
HAY-PID-08005	08005-: 01, 02, 03

8 Missing Information / Cannot Be Verified

- **Sizes removed or never marked:** high-point vents on 14" AG lines (02013-05/02115-05), KOD-top valves (02013-07/02115-07), side-stream loop sizes (02015B — 1" labels erased on current set), FV-1240A manifold intent (03002-05), anti-foam "inst..." fragment (08005-04).
- **Line references incomplete:** slop-tank drain line 2 number partially legible; one tie-in P&ID reference blank (04001-08/-09).
- **Ratings/materials for added small-bore items:** not stated — order per governing line class; Verify in Piping Material Specification.
- **Closure basis:** Rev 1 closes items on markup absence only; no revision letters changed between sets, and the base drawings were not updated to incorporate the removed marks. Physical completion of closed items is not verified by this review.

9 QC / Verification Statement

- Both markup sets (Rev 0 source and current authoritative set) rendered at 150 DPI with 300-600 DPI re-renders for every uncertain mark; all 40 sheets re-read in full.
- Every Rev 0 item received an individual KEEP/DROP verdict with location-level justification; zero UNSURE verdicts remained.
- Title blocks compared on all 40 sheets: no revision changes between sets — deletions are markup cleanup, not drawing re-issues.
- Duplicates: three cross-sheet pairs merged with full traceability; no item key reused; disposition of all 295 closed/merged keys recorded in Section 7.
- BOM and procurement rebuilt strictly from the 137 open items; Rev 0 material lines for closed items withdrawn.

Post-issue verification record (2026-08-12) — three independent check batteries, all passed

- **Battery 1 — verdict reconciliation:** the per-page verdict files written independently by the eight verification reviewers were re-parsed and compared against this document: 430/430 items covered (138 KEEP / 292 DROP / 0 UNSURE), zero conflicts between files, and zero discrepancies against the open list in either direction. Open-item arithmetic confirmed: 138 — 3 merged + 2 new = 137.
- **Battery 2 — deliverable internals:** all sheet-section counts match their rows; zero duplicate item keys; summary tiles match counted rows (137 / 15 critical / 292 closed / 3 merged / 2 new / 38 BOM / 10 questions); Disposition Register totals 292 + 3 with zero overlap against open items and closed + merged + carried-over = 430; every BOM and engineering-question reference resolves to an open item; workbook master matches this report field-for-field on all 137 rows; discipline tabs union to 137 with critical-first sorting; Disposition sheet carries 295 rows incl. 16 SAFETY and 3 MERGED tags; PDF contains all 137 keys; repository copies byte-identical (SHA-256) to issued files.
- **Battery 3 — physical evidence:** an automated red/pink mark-cluster scan of all 40 sheets of the current set reconciles with the open list on every page. Explained variances: HAY-PID-04001 items are x4-tank multi-instance rows (~15 physical marks under 6 items); 01001-33 is a graphite (pencil) mark invisible to the red detector; the three adjacent HAY-PID-08005 items merge into one cluster.
- **Closure basis restated:** Rev 1 closures rest on mark absence from the current markup only — physical completion of closed items is not certified; the 16 SAFETY-tagged dispositions require PSSR walkdown confirmation.

Rev 1 — corrected 2026-08-12 against the current authoritative redline set ("mplx_redline.pdf", 40 sheets); supersedes Rev 0 (2026-08-09, 430 items). Item keys are continuous across revisions; closed items retain their keys in the Disposition Register. This document consolidates field redlines for construction action, procurement, and engineering disposition; it does not supersede issued-for-construction drawings, and closure of a punch item here does not certify physical completion.